UNIVERSITY OF KwaZulu-Natal
School of Information Systems & Technology

ISTN 3AS Computer Programming Supplementary Test
Applied Systems Analysis (ISTN3AS)

DURATION: ***1 Hour*** **(add 15 minutes reading time + 15 minutes submission)**

TOTAL MARKS: 70 + ***5 Marks for correct and timeous submission***

INTERNAL EXAMINERS: Dr S Ranjeeth and Dr D Kumar

INSTRUCTIONS TO CANDIDATES:

1. The questions below are to be answered using VS 2022 and SQL Server
2. This examination paper consists of **5 numbered pages** including the cover page. Ensure that you have all pages.
3. Project Creation...

Open Visual Studio 2022 and create a new **C# Windows Forms** project named according to the following convention:

First 3 letters of your surname + _Student Number
e.g. **Khu_123456789**

The project should be **saved on the desktop and once you have completed the solution, Zip up the entire solution folder, copy and paste (upload) the zipped version into the Submission folder on Learn 2026.**
Any late submission will incur a penalty
Note: *You are expected to save your project at regular intervals*

<u>Submission</u>: If you have finished your solution early and you have made the submission, then you can ask the invigilator to verify that your solution has been submitted

<u>Sql Server: 146.230.177.46\ISTN3</u>

<u>Main System Requirement</u>

You are required to create a database driven front-end application for a **Book retail shop**. The objective of the application is to:

- **Display details** of all books that are available for purchase
- Provide a facility whereby a **search function** displays all details of those book titles/names that match the first few letters of the "search text"
- Create a **Purchase invoice** that displays the details of all books that are being purchased and the overall total amount owing
- Record the *overall* sale transaction (not the inner detail)

System Specifications and Mark Allocation

PART ONE – DB setup (10 minutes, 10 marks)

1. Establish a connection to the SQL Server database using your personal password and login. Create **2** tables named “***Book***” and “***BookOrder***”.
   The Design of the 2 tables are illustrated below

(5 marks)

MAJOR\MAJOR.sanjayr - dbo.Book* × MAJOR\MAJOR.sanjayr - dbo.Book

Column Name	Data Type	Allow Nulls
BookTitle	varchar(50)	☐
BookCost	decimal(8, 2)	☐
QOH	int	☐

MAJOR\MAJOR.sa...r - dbo.Table_1* × MAJOR\MAJOR.sanjayr - dbo.Book*

Column Name	Data Type	Allow Nulls
OrderNumber	int	☐
OrderDate	date	☐
OrderAmt	decimal(8, 2)	☑

Column Properties

Identity Specification	Yes
(Is Identity)	Yes
Identity Increment	1
Identity Seed	1

2. By making use of the Sql Server client interface, add the following records into the table named ***Book***.

MAJOR\MAJOR.sanjayr - dbo.Book × MAJOR\MAJOR.sa...r - dl

BookTitle	BookCost	QOH
ASP3	975.00	23
C# is CooL	605.00	12
C# Journey	490.00	20
DB in C#	675.00	3
EF is Cool	675.00	10
Java1	795.00	17
Java2	575.00	12
Java3	755.00	2

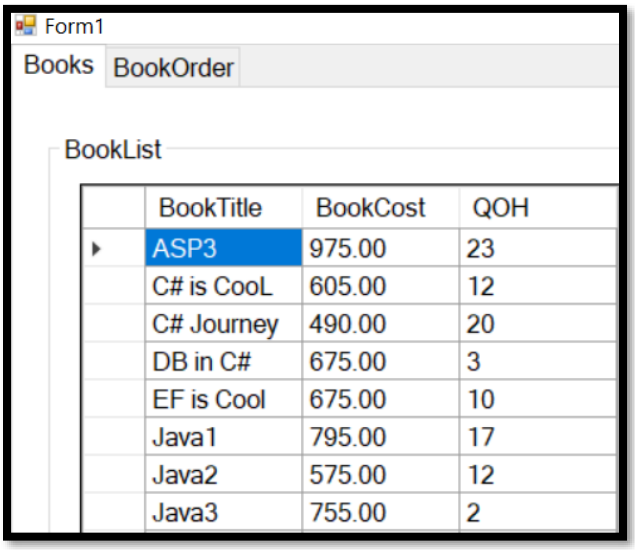
(5 marks)

PART TWO

Windows Forms App (C#) *Menu & Display* (10 minutes, 8 marks)

3. Create a user interface (UI) that consists of a form with a **Tab control** that is docked (choose the fill option) with 2 tabs as shown below:

4. On the first tab (named Books), add a GroupBox with the display text "*BookList*" and **display a list of all Books** from the table named Books using a datagridview

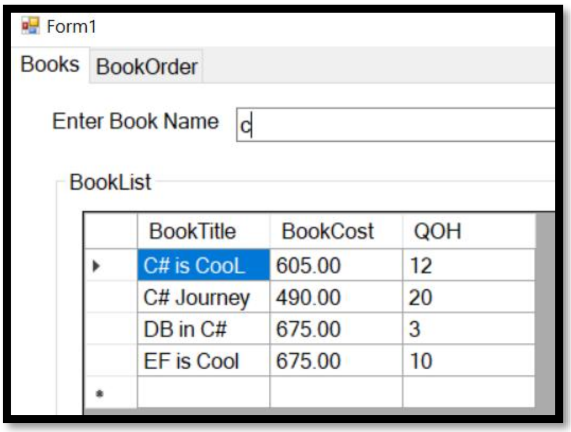


(8 marks)

PART THREE

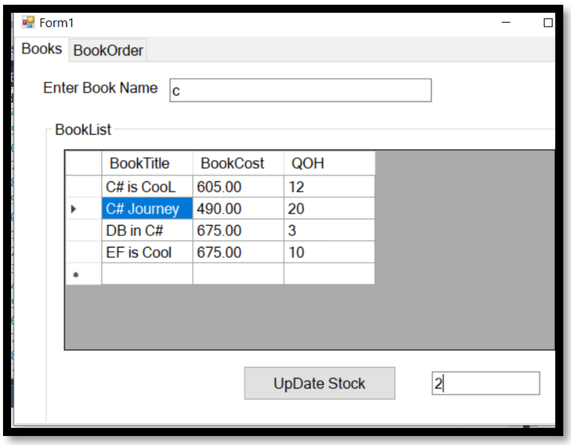
Windows Forms App Dynamic Search & Update (20 minutes, 20 marks)

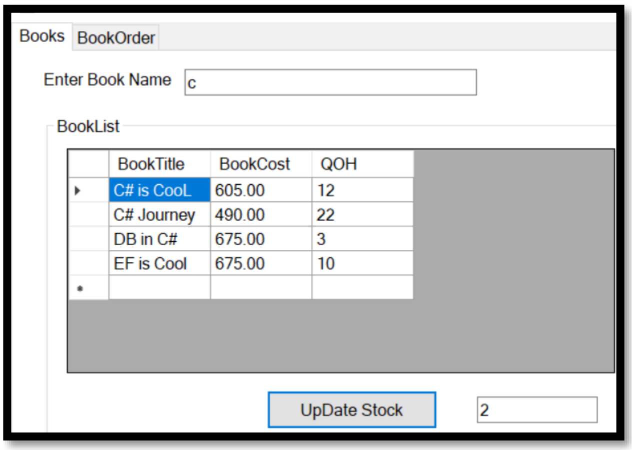
5. Add functionality to the interface so that the user is able to perform a ***dynamic search for a book name*** that is entered via a textbox as shown below.



(10 marks)

6. Add functionality to the interface so that the user has an opportunity to enter an amount by which the stock quantity on hand (**QOH**) will increase or decrease for the current row (i.e. the row in which the cursor is placed). A **positive amount** will result in the **stock increasing** by that amount. A **negative amount** will result in a **decrease** in the stock QOH. The change in stock amount is "triggered" by the *click of a button* as illustrated below:





Once the update of stock is performed, the new stock value is reflected in the Gridview

Points to ponder: pay special attention to data types;

dataGridviewBook.CurrentRow.Cells[…].Value.ToString()   // may be useful to know

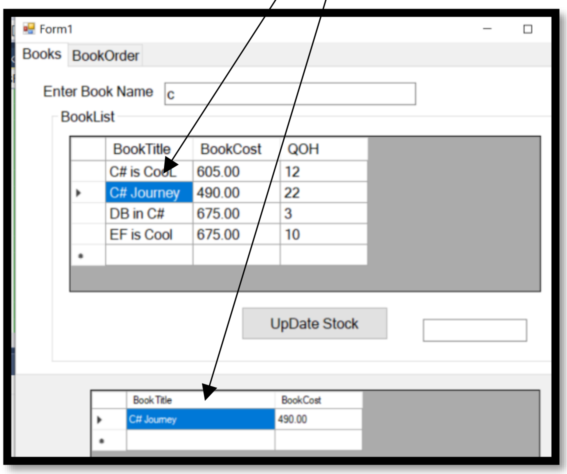
You have now passed the Test with 54%! ☺

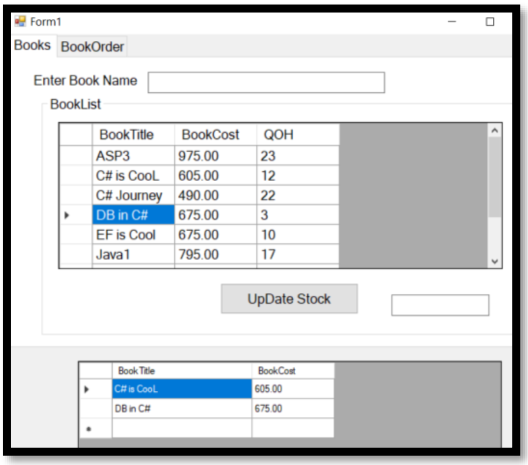
(10 Marks)

PART FOUR

Windows Forms App Generate INVOICE Grid (20 minutes, 12 marks)

7. Once a ***cell content*** is double clicked (the event handler is ***CellContentDoubleClick***), **add that row** to an invoice data table and display this invoice **on a new data grid** (the current tab page).



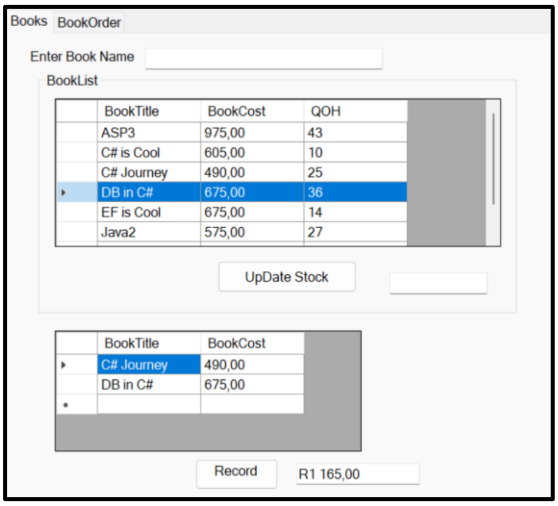


(12 marks)

PART FIVE

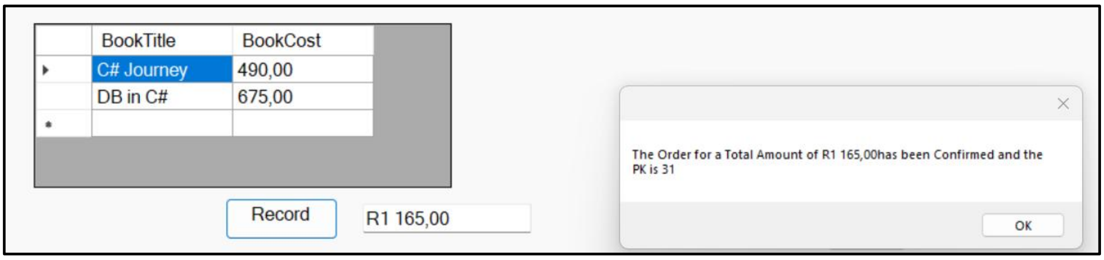
Windows Forms App Generate TOTAL SALE AMOUNT (20 minutes, 20 marks)

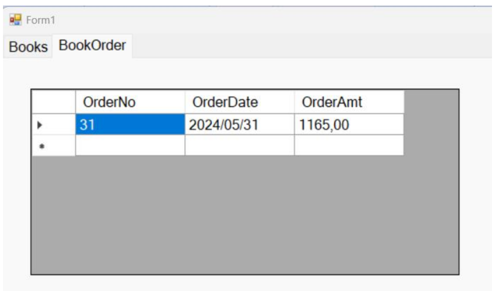
8. Display ***the total purchase amount owing*** for the invoice. This value updates each time a new row is added to the Invoice.



(8 marks)

9. Add a button that enables you to record the sale into the Book Order table created in Question One. The only information that will be recorded is the total sale amount and the current date. Once a sale has been recorded, provide a **confirmation message informing the user of the sale detail** as well as the **primary key of the newly recorded sale**. On the 2nd tab named BookOrder, add a data gridview that will present an updated view of all the orders/sales conducted.





The OrderNo value is auto-incremented; the number shown in the image is just for display and may be different on your app

(12 marks)

<END OF PAPER>